Personal Budget Tracker
Database Design Document

CP476 - Internet Computing

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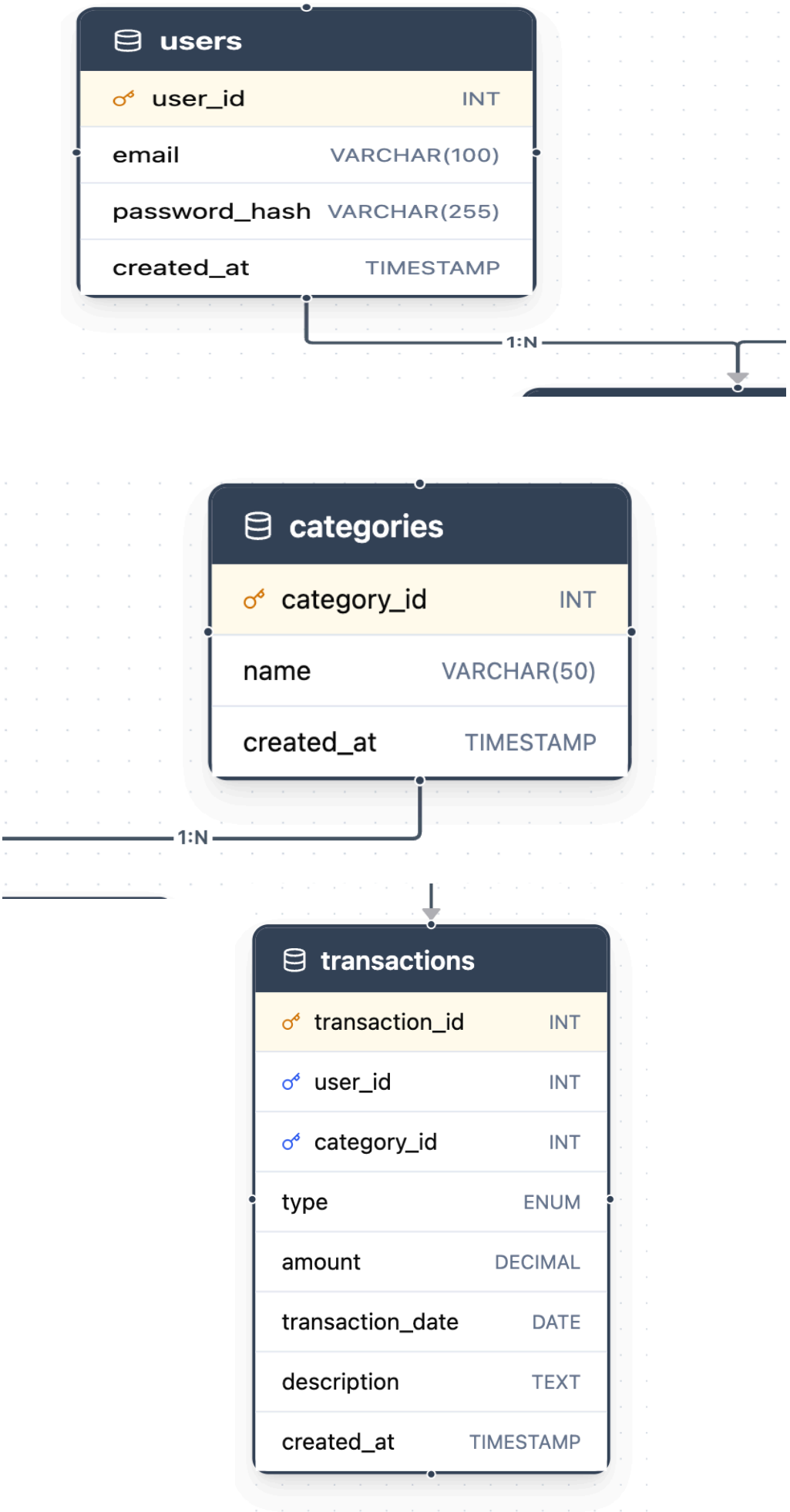
users
user_id INT
email VARCHAR(100)
password_hash VARCHAR(255)
created_at TIMESTAMP
1:N
categories
category_id INT
name VARCHAR(50)
created_at TIMESTAMP
1:N
transactions
transaction_id INT
user_id INT
category_id INT
type ENUM
amount DECIMAL
transaction_date DATE
description TEXT
created_at TIMESTAMP

users
user_id INT
email VARCHAR(100)
password_hash VARCHAR(255)
created_at TIMESTAMP
categories
category_id INT
name VARCHAR(50)
created_at TIMESTAMP
1:N
1:N
transactions
transaction_id INT
user_id INT
category_id INT
type ENUM
amount DECIMAL
transaction_date DATE
description TEXT
created_at TIMESTAMP

SQL CREATE TABLE Statements

```
CREATE TABLE users (
  user_id INT AUTO_INCREMENT PRIMARY KEY,
  email VARCHAR(255) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

```
CREATE TABLE categories (
  category_id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(80) NOT NULL UNIQUE,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

```
CREATE TABLE transactions (
  transaction_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  category_id INT NOT NULL,
  type ENUM('income','expense') NOT NULL,
  amount DECIMAL(10,2) NOT NULL,
  description VARCHAR(255) NULL,
  transaction_date DATE NOT NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT fk_transactions_user
    FOREIGN KEY (user_id) REFERENCES users(user_id)
    ON DELETE CASCADE,

  CONSTRAINT fk_transactions_category
    FOREIGN KEY (category_id) REFERENCES categories(category_id)
    ON DELETE RESTRICT,

  CONSTRAINT chk_amount_positive
    CHECK (amount > 0)
);
```